

WEEK 2

Project Title : Stationery Shop Management System

Database Description

The **Stationery Shop Management System** is designed to manage all activities of a stationery shop such as customer details, product information, suppliers, orders, billing, inventory, payments, and employees. The database stores data in multiple related tables to reduce redundancy and improve efficiency. Each entity has a unique primary key, and relationships ensure data integrity between tables.

Entities, Attributes & Primary Keys

Entity	Attributes	Primary Key
Customer	Customer_ID, Customer_Name, Phone_No, Email, Address	Customer_ID
Employee	Employee_ID, Employee_Name, Phone_No, Position, Salary	Employee_ID
Supplier	Supplier_ID, Supplier_Name, Phone_No, Email, Address	Supplier_ID
Category	Category_ID, Category_Name	Category_ID
Product	Product_ID, Product_Name, Brand, Price, Stock_Quantity, Category_ID, Supplier_ID	Product_ID
Order	Order_ID, Customer_ID, Employee_ID, Order_Date, Total_Amount, Order_Status	Order_ID
Order_Item	Order_Item_ID, Order_ID, Product_ID, Quantity, Unit_Price	Order_Item_ID
Payment	Payment_ID, Order_ID, Payment_Date, Payment_Method, Amount	Payment_ID
Inventory	Inventory_ID, Product_ID, Available_Stock, Reorder_Level, Last_Updated	Inventory_ID

Relationships

Relationship	Cardinality
Customer → Order	One Customer places Many Orders (1:M)
Employee → Order	One Employee handles Many Orders (1:M)
Supplier → Product	One Supplier supplies Many Products (1:M)
Category → Product	One Category contains Many Products (1:M)
Order → Order_Item	One Order has Many Order Items (1:M)
Product → Order_Item	One Product appears in Many Order Items (1:M)
Product → Inventory	One Product has One Inventory Record (1:1)
Order → Payment	One Order has One Payment (1:1)

Foreign Keys

Table	Foreign Key
Product	Category_ID, Supplier_ID
Order	Customer_ID, Employee_ID
Order_Item	Order_ID, Product_ID
Payment	Order_ID
Inventory	Product_ID

STATIONERY SHOP MANAGEMENT SYSTEM - ER DIAGRAM

